Graph Sketching: Inverse Hyperbolic Functions

Questions 1–12

Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

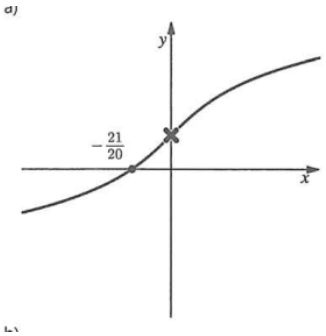
a) $y = \operatorname{arsinh} x$
b) $y = \operatorname{arcosh} x$
c) $y = \operatorname{artanh} x$

Question 2. Sketch the following pairs of graphs on the same axes. Label each graph clearly and show where they intersect the coordinate axes:

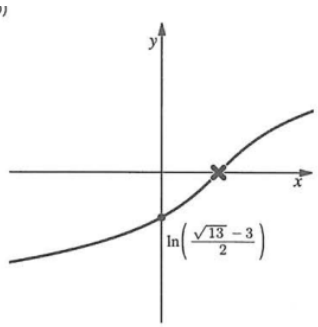
a) $y = \operatorname{arsinh} x$ and $y = 3\operatorname{arsinh} x$
b) $y = \operatorname{arsinh} x$ and $y = \operatorname{arsinh}(x - 2)$
c) $y = \operatorname{arsinh} x$ and $y = -\operatorname{arsinh} x$
d) $y = \operatorname{arcosh} x$ and $y = \operatorname{arcosh} \frac{1}{2}x$
e) $y = \operatorname{arcosh} x$ and $y = \operatorname{arcosh}(x + 3)$
f) $y = \operatorname{arcosh} x$ and $y = \operatorname{arcosh}(-x)$

Question 3. By considering the point of inflection (identified on the graphs by a ×), find the equation of each of the following graphs in the form $y = \operatorname{arsinh}(x - a)$ or $y = \operatorname{arsinh} x + b$, where a and b are integers:

a)

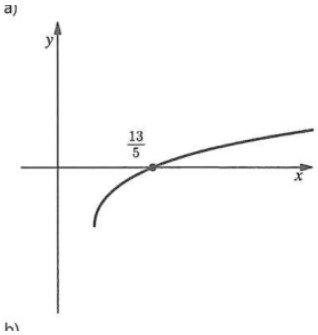


b)

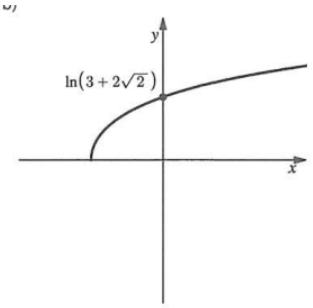


Question 4. Find the equation of each of the following graphs in the form $y = \operatorname{arcosh}(x - a)$ or $y = \operatorname{arcosh} x + b$, where a and b are integers:

a)



b)



Question 5. State the equations of the asymptotes of the following graphs:

a) $y = \operatorname{artanh}(x + 5) - 3$

b) $y = \operatorname{artanh}(x - \frac{1}{3}) + 2$

Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \operatorname{artanh} x - 1$

b) $y = \operatorname{artanh}(x + 3)$

c) $y = \operatorname{artanh}(x - \frac{1}{2})$

d) $y = -\operatorname{artanh} x$

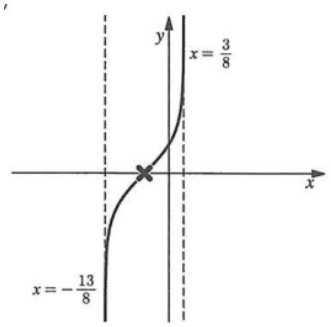
Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \operatorname{artanh} 3x$

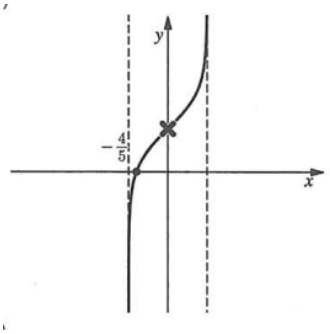
b) $y = \operatorname{artanh} \frac{1}{4}x$

Question 8. In each of the following graphs, the point of inflection is identified by a $\times$. Find the equation of each in the form $y = \operatorname{artanh}(x - a)$ or $y = \operatorname{artanh} x + b$, where a and b are integers:

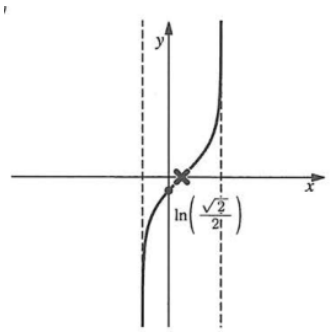
a)



b)



c)



Question 9. Sketch the following graphs, given $k > 1$, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \operatorname{arsinh}(x - k) + k$
b) $y = \operatorname{arcosh}(x + k) - k$
c) $y = \operatorname{artanh}(x - k) - k$

Question 10. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

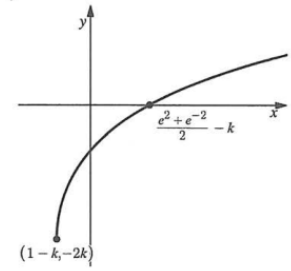
a) $y = 2 - \operatorname{arsinh} x$
b) $y = -\operatorname{arcosh} 2x$
c) $y = \operatorname{artanh}(\frac{1}{2}x + 1)$
d) $y = \operatorname{arsinh}(1 - x)$
e) $y = -\operatorname{arcosh}(-x)$
f) $y = -\operatorname{artanh} 2x$

Question 11. Both of the following inverse hyperbolic curves are transformations of $y = \operatorname{arcosh} x$. No horizontal stretches have been performed. Given k is a constant where $1 < k < 3$, find a suitable equation for each:

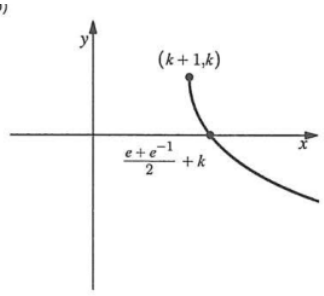
a)

suitable equation for each:

a)



b)



Sketch the following graphs, given $k > 1$

Question 12. Sketch the following graphs, given $k > 1$, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = k - \operatorname{arsinh}(x + k)$
b) $y = \operatorname{arcosh}(x - 3k)$
c) $y = k - \operatorname{artanh}(kx + k)$